Tutorial Week 1

Pre-reading/viewing: Lecture 1, document '**Software required for this module'**, document '**Using AppsAnywhere guid**e'

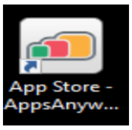
The tutorial covers the following.

1. Launching Python/IDLE on the lab PCs.
2. Experimenting with the Python Shell
3. Running programs in Script Mode - Arithmetic Operators & printing strings

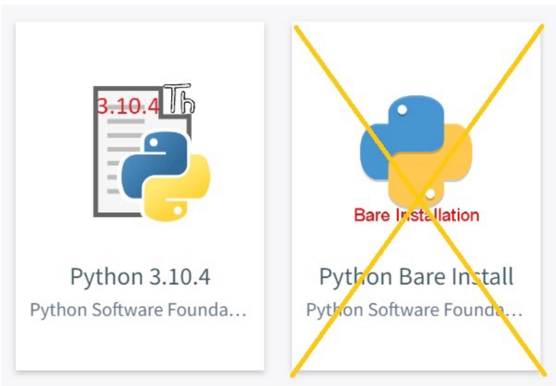
3 Order of operations
4 New
5 New

1: Launching Python/IDLE on the lab PCs

- The lab PC's have Python 3.10.4 (includes IDLE) available via AppsAnywhere. When logged in, click on the AppsAnywhere icon:



- Then scroll down or type 'IDLE' or 'Python' in the **Search Apps** box.
- Launch version **Python 3.10.4 (NOT BARE INSTALLATION)**
- See document '**Using AppsAnywhere guide**' for further details.

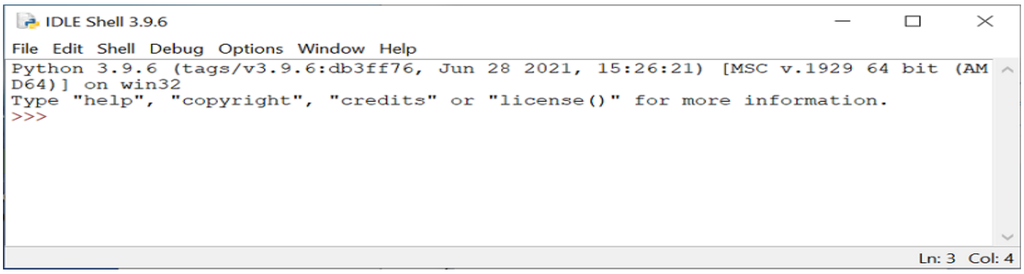


Accessing Python from home.

- See 'Guide to Using AppsAnywhere on Your Personal Computer'.
- Mac Users - AppsAnywhere does not work with Macs.
- Download Python from: https://www.python.org/downloads/
- Any Python 3.10.x (or Python 3.9) version will be suitable.
- Or, for the first tutorial use an 'online Python 3 compiler'. E.g., https://www.programiz.com/python-programming/online-compiler/

2: Experimenting with the Python Shell

We can use Python in two modes - Python Shell and Script Mode. We will experiment with using the python shell. When you launch Python the Python Shell window will display as follows.



- The **Shell window** allows you to type Python code at the prompt string and see the result immediately. It is known as **interactive mode**.
- >>> is the prompt string.
- In the Python shell, we could enter the following calculations line by line and hit enter to get the result.

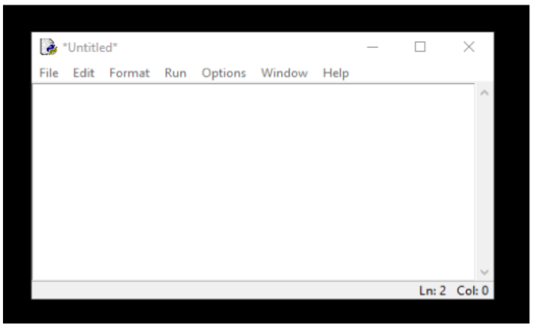
```
>>> 88 + 4
92
>>> 45 * 4
180
>>> 17 / 3
5.666666666666667
```

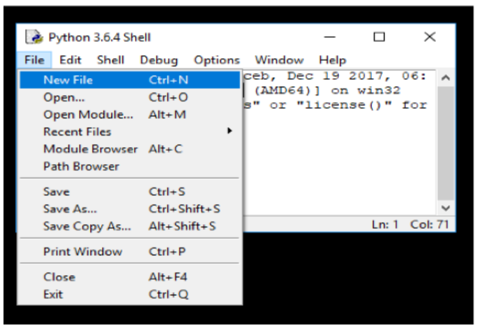
- Try the above in the Python shell.
- Python Shell is good for testing small amounts of code however the statements are not saved anywhere. If you want to execute a set of statements multiple times use script mode.

3: Run programs in Script Mode

The interactive Shell window allows us to open a program editing window.

• From the Python Shell window, select New File from the File menu. You will see a window entitled "Untitled".





- From the **File** menu, select **Save As**, and select a folder to save your Python program file.
- On the University PCs you need to **save to your H drive.** In the **Filename:** textbox, type: test1.py. Then click on the **Save** button.
- You will then see a blank editor window ready for you to type in the following Python programs.

a) Arithmetic Operators

- Write down the output of the following program.
- Then create, save and run the program.
- Check that you get the right values for all of them?

```
num = 5
print(num)
num = num + 2
print(num)
num = num // 3 * 6
print(num)
print(7 + 15 % 4)
num = 24 // 3 // 4
print(num)
```

b) Printing strings

i. Create, save and run the following program.

```
a = "Hello out there."
print(a)
b = 'Hello again.'
print(b)
```

Modify the program so that it concatenates (joins) variables a and b and prints the result.

ii. Why would the following produce an error?

```
total = 10
greet = 'Hello'
both = total + greet
print(both)
```

- Note that you cannot concatenate a string and an integer.
- Use the built-in str() function (converts to string) to fix the error.

iii. What will be displayed by the following code? Type and run the code to check your answer.

```
print("A", end = ' ')
print("B", end = ' ')
print("C", end = ' ')
print("D", end = ' ')
```

iv. What is the output of the following program? Test the program to check your answer.

```
a = '10'
b = '99'
c=a+b
print(c)
```

• Modify the above program so that it prints out the value **109**

v. Write the code using formatted string literals (f-strings) to format two variables **a** and **b** such that they are printed to the Shell as show below (for values a=24 and b=17)

```
First variable is currently 24. The second variable is currently 17
```

c. Order of Operators

i. The Python code below runs without error but the expected print output was **145**
re-write and run the code so it outputs the correct result

```
a=17
b=12
c=5
result= a+b*c
print (f'The result is {result}')
```

The code examples below all give an incorrect output
Re-write and run the code examples below so they output the correct result

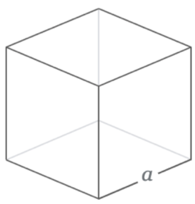
ii.

```
#find the average of three numbers
red=24
green=12
blue=63
greyscale_average = red + green + blue / 3
print (f'The greyscale value is {greyscale_average}')
```

expected result: 33.0

iii.

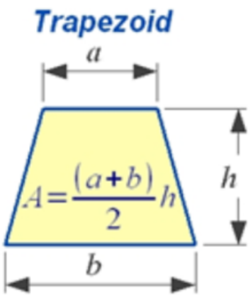
$$V = a^3$$




```
#find the total volume of two cubes
cube_1_side=11
cube_2_side=4
total_volume=(cube_1_side+cube_2_side)**3
print (f'The  total volume of both cubes is {total_volume}')
```

expected result: 1395

iv

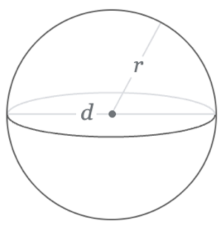



```
#find the total area of a trapezoid
a=10
b=12
h=7
Area=a+b*h/2
print (f'The area of the trapezoid is {Area}')
```

expected result: 77.0

v.

$$V = \frac{4}{3}\pi r^3$$




```
#find the total volume of a sphere from the radius
sphere_radius=2
sphere_vol= 4/3 * (3.142*sphere_radius)**3
print (f'The volume of the sphere is {sphere_vol}')
```

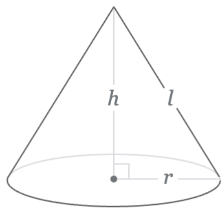
expected result: 129.9461627238613

Additional Exercises

Work in Groups Pairs to fix and execute the code below

The code below is to be used to calculate the volume V in litres of a right circular cone for a given variable height=3m and radius=2m

$$V = \pi r^2 \frac{h}{3}$$



However, when the code executes it displays a number of errors. Analyse the code and re-write it so that it executes, calculates the correct results and formats the print to screen as shown

```
1_height_m=3
2_radius_m=2
pi=3.142

Volume in cubicMetres=(pi*2_radius_m)**2*1_height_m/3

print ("The volume of the cone is "+Volumeinm3+"cubic metres")
Volume in litres= Volume in cubicMetres*1000
print ("or "+Volumeinlitres+"ltrs.")
```

Expected output:
`The volume of the cone is 12.568 cubic metres or 12568.0 ltrs.`